

Mengxin Xie

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EDUCATION

Imperial College London

London, UK

MSc Risk Management & Financial Engineering

September 2023 – December 2024

- Core Courses: Stochastic Calculus, Financial Statistics, Advanced Option Pricing, Computational Finance with C++, Python Data Structures and Algorithms, Portfolio Management and Asset Allocation, Risk Management

The Hong Kong Polytechnic University

Hong Kong, China

BSc (HONS) Investment Science & Finance Analytics

September 2019 – May 2023

- Awards: First Class Honours, Dean's Honours List
- Core Courses: Machine Learning, Time Series Analysis, High-Dimensional Data Analysis, Derivative Pricing

PROFESSIONAL EXPERIENCE

Chicago Mercantile Exchange (CME) Group Inc

London, UK

Quantitative Risk Research Analyst

July 2024 – Present

- Forward Curve Construction:** Built 10 energy forward curves for products with limited historical data using proxy interpolation, CoC borrowing, and simulation methods. Improved curve data quality for 20 agriculture products through data quality check, stale contract price filtering, and enhanced extrapolation methodologies.
- SPAN2 Parameter Calibration and Margin Backtesting:** Calibrated model parameters (e.g., hypothetical scenarios, time-to-maturity scalars); backtested trading strategies and provided margin-level explanations.
- New Product Launch & Migration:** Supported SPAN2 checklist preparation and VT verification; updated C#Library and RiskLibrary configurations for new products; collaborated with QQA/TRE teams to conduct QA/PP environment testing and resolved margin-number mismatches; developed automation tools to streamline processes.
- C# Library Enhancement:** Expanded functionality of internal C# libraries to support methodology updates; implemented bug fixes and unit tests; diagnosed issues in C# daily curve-generation pipelines and deployed hotfixes to maintain system stability.

Deloitte

Shanghai, China

Quantitative Modelling Intern

June 2023 – August 2023

- Constructed features from large-scale fundamental datasets and applied various techniques, including Synthetic Minority Over-sampling Technique (SMOTE) algorithm to address class imbalance problem.
- Developed a predictive XGBoost model to assess probability of equities receiving delisting warnings on the Shanghai Stock Exchange, achieving 88.8% out-of-sample recall in identifying high-risk equities.

Castalia Capital Management

Beijing, China

Quantitative Research Intern

June 2022 – August 2022

- Predicted the volatility of ETF daily returns with ARMA-GARCH model and applied volatility scaling techniques to adjust position sizes in a momentum trading strategy, improving strategy's Sharpe ratio by 13%.
- Predicted stock returns using LightGBM with fundamental and technical indicators; enhanced model performance and robustness via rolling-window cross-validation and hyperparameter optimization.

PROJECT EXPERIENCE

American Option Pricing with Monte Carlo Simulation:

- Implemented Least Squares Monte Carlo, Weighted Least Squares Monte Carlo, and Parallel Monte Carlo methods to price strike-fixed American Put Options with varying initial prices, volatilities, and expiration times.
- Benchmarked results against Finite Difference Method to compare algorithms' pricing speed and accuracy, and calibrated market data on the SPDR S&P 500 ETF (SPY) American put options.

Market Regimes Identification with Clustering:

- Adopted PCA and Gaussian Mixture Model clustering algorithm to identify distinct market regimes from macroeconomic indicators.
- Experimented with algorithms including Linear Regression, Lasso Regression, Random Forest to select the best stock return prediction model for each regime.
- Integrated best-performing models into a predictive framework to continuously monitor and rebalance an equally weighted investment portfolio from the S&P 500 stock pool monthly, achieving better backtest results than baseline model.

ADDITIONAL SKILLS

- Programming Skills:** Python, C++/C#, SQL
- Language Skills:** English, Mandarin
- Certificate:** CFA Level I (passed)